

Exploring the Security Problems with Malware and Ransomware

Carson Blin

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1 Introduction

The topic of interest I want to tackle in this semester-long project is regarding malware and ransomware security in our everyday devices and how to protect against such vulnerabilities. As time progresses, technology is going to get better and smarter, but from a security standpoint can remain static or outdated until a threat actor is introduced and starts infecting devices at a greater scale, leading to many users losing data and businesses suffering great financial loss.

2 Reviewed Literature

Throughout this literature review process, I have been exposed to many concepts, knowledge, and findings that will help me synthesis an interesting and thought-provoking report regarding malware security. The findings regarding malware and ransomware are below:

- Ibrar Yaqoob et al., “The rise of ransomware and emerging security challenges in the Internet of Things,” Special Issue on 5G Wireless Networks for IoT and Body Sensors, *Computer Networks* 129 (2017): 444–458, ISSN: 1389-1286, <https://doi.org/10.1016/j.comnet.2017.09.003>, <https://www.sciencedirect.com/science/article/pii/S1389128617303468>
- T.R. Reshmi, “Information security breaches due to ransomware attacks - a systematic literature review,” *International Journal of Information Management Data Insights* 1, no. 2 (2021): 100013, ISSN: 2667-0968, <https://doi.org/10.1016/j.jjime.2021.100013>, <https://www.sciencedirect.com/science/article/pii/S2667096821000069>
- Noor Thamer and Raaid Alubady, “A Survey of Ransomware Attacks for Healthcare Systems: Risks, Challenges, Solutions and Opportunity of Research,” in *2021 1st Babylon International Conference on Information Technology and Science (BICITS)* (2021), 210–216, <https://doi.org/10.1109/BICITS51482.2021.9509877>

- Sarah Tawfiq Albassam et al., “Ransomware Detection in the Internet of Things (IoT): Challenges and Emerging Solutions,” in *2024 Seventh International Women in Data Science Conference at Prince Sultan University (WiDS PSU)* (2024), 183–190, <https://doi.org/10.1109/WiDS-PSU61003.2024.00046>
- Syed Rameem Zahra and Mohammad Ahsan Chishti, “RansomWare and Internet of Things: A New Security Nightmare,” in *2019 9th International Conference on Cloud Computing, Data Science Engineering (Confluence)* (2019), 551–555, <https://doi.org/10.1109/CONFLUENCE.2019.8776926>

3 Summary of Learning

After reading the research papers listed above, I have been exposed to what specific malware is most common in the business world. From a business perspective, its important to mitigate ransomware attacks to avoid financial loss and creating a bad image for the company. Ransomware is not limited to the tech industry, but many industries that heavily rely on technology to operate like the medcial industry. The medical industry is a prime target for threat actors due to the amount of user data in their possession. A few common types of ransomware are crypto ransomware, botnets, Locker ransomware, ransomware-as-a-service (RaaS), hybrid ransomware, and social engineering tactics. These attacks utilize encryption to encrypt the user’s data and ask for money to have access to the encrypted data. These types of ransomware attacks are used by threat actors to infect and compromise devices all across the globe. A lot of these devices are commonly used devices that belong under the Internet of Things (IoT) umbrella such as watches, smart trackers, smart sensors, and any smart device that is able to send and receive data from the Internet. Many of these ransomware attacks are being deployed through compromised websites and accounts, downloads, and phishing. A few popular methods to prevent ransomware are through machine learning (ML) technologies and blockchain technology. The current problems regarding these technologies are can be scalability and high costs.

4 Next-Phase Plan

In my next report, my next few objectives are to strategically plan out effective methods for this project, begin the data collection stage, and conduct another literature review to ensure the content and knowledge are correct and accurate. Another important step is to refine my topic for the project regarding security measures for ransomware attacks.